

BLUETOOTH CONTROLLED ROBOT CAR WITH OBSTACLE AVOIDANCE

A Mini Project

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ABSTRACT

The Bluetooth Controlled Robotic Car is an embedded system project designed to enable wireless control of a robotic vehicle using Bluetooth communication technology. The system uses a smartphone application to send control commands to an Arduino microcontroller through a Bluetooth module. Based on the received commands, the Arduino processes the input signals and controls the movement of Direct Current motors through a motor driver module. The robotic car can perform basic movements such as forward, backward, left turn, right turn, and stop. The system integrates both hardware and software components including Arduino Uno, Bluetooth module, motor driver, Direct Current motors, and battery power supply. The primary objective of this project is to demonstrate wireless communication, motor control, and real-time embedded system operation. The developed system provides a simple and efficient method for remote vehicle control and can be further extended for applications such as surveillance robots, automated vehicles, and smart robotic systems.

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LIST OF ABBREVIATIONS

ASCII	American Standard Code for Information Interchange
DC	Direct Current
HC-SR04	Hardware Components - Sensor Range 04
IDE	Integrated Development Environment
MCU	Microcontroller Unit
PCB	Printed Circuit Board
SPP	Serial Port Profile
UART	Universal Asynchronous Receiver-Transmitter
USB	Universal Serial Bus
UML	Unified Model Language

CHAPTER 1

INTRODUCTION

This chapter provides an overview of the Bluetooth Controlled Car with Obstacle Avoidance project and its importance in the field of embedded and robotic systems. It explains the concept of wireless control using Bluetooth communication and highlights how mobile devices can be used to control robotic vehicles in real time. The chapter also describes the integration of sensors for obstacle detection to improve safety and automation. Additionally, it presents the main objectives of the project and briefly explains the system architecture and methodology used for designing and implementing the robotic car. This chapter serves as a foundation for understanding the design, development, and testing phases discussed in the subsequent sections of the report.

1.1 OBJECTIVE OF THIS PROJECT

The primary objective of this project is to design and implement a Bluetooth Controlled Robotic Car with Obstacle Avoidance using an embedded system approach. The project aims to enable wireless control of a robotic vehicle through a smartphone application using Bluetooth communication. It focuses on integrating a microcontroller, motor driver, and Direct Current (DC) motors to achieve accurate movement control in different directions. Additionally, the system incorporates an ultrasonic sensor to detect obstacles and improve operational safety by preventing collisions. The project also aims to demonstrate the practical implementation of wireless communication, real-time control, and sensor-based automation in robotics, providing a low-cost and efficient platform for learning embedded systems and intelligent vehicle control.

1.2 PROJECT OVERVIEW

- This project focuses on developing a **Bluetooth Controlled Robotic Car with Obstacle Avoidance**, enabling wireless control using a smartphone. Unlike traditional wired systems, it provides flexible and real-time vehicle operation through Bluetooth communication.
- The system integrates components such as an Arduino microcontroller, Bluetooth module, motor driver, Direct Current (DC) motors, and an ultrasonic sensor. The Arduino processes commands received via Bluetooth and controls the motors, while the ultrasonic sensor detects obstacles and ensures safe movement.
- The project follows a structured approach involving command transmission, signal processing, motor control, and obstacle detection. It results in a reliable robotic system that demonstrates the practical use of embedded systems, wireless communication, and basic automation in robotics.

1.3 ORGANIZATION OF THE REPORT

Chapter 1

Presents an introduction to the project, project background, objectives, and an overview of the system and its purpose.

Chapter 2

Includes the system analysis, describing the existing system, proposed solution, project requirements, hardware/software specifications, and tools used.

Chapter 3

Explains the system architecture and design, detailing the structural components and functional workflow of the proposed model.

Chapter 4

Describes the implementation phase, the development process, testing methods, challenges faced, and the results obtained.

Chapter 5

Concludes the report by summarizing key outcomes, highlighting summary, and suggesting future enhancements to improve the system.

CHAPTER 2

SYSTEM ANALYSIS

System Analysis involves examining and understanding the existing network monitoring methods and identifying limitations in traditional approaches. The objective is to determine operational gaps and design an improved solution that enables real-time device discovery, profiling, and monitoring with greater accuracy and automation. This analysis breaks down system components and evaluates how they interact to achieve efficient network management.

2.1 EXISTING SYSTEM

In traditional robotic vehicle systems, control is often achieved through wired connections or simple remote controllers that operate using radio frequency signals. These systems generally require direct manual control and do not provide intelligent features such as obstacle detection or automated safety mechanisms. As a result, the vehicle may collide with objects if the operator does not respond quickly. Additionally, many basic robotic cars lack wireless smartphone integration and rely on limited control interfaces, reducing flexibility and user convenience. The absence of sensor-based automation also limits the system's ability to respond dynamically to environmental conditions. Consequently, existing systems suffer from restricted mobility, reduced safety, limited scalability, and minimal automation capabilities.

2.2 PROPOSED SYSTEM

The proposed system is a Bluetooth Controlled Robotic Car with Obstacle Avoidance that enables wireless control and improved safety. A smartphone application sends movement commands via Bluetooth, which are received by the Arduino microcontroller to control the motors. An ultrasonic sensor continuously detects obstacles and automatically stops the robot when an object is within a set distance, ensuring collision prevention. This integration of wireless communication and sensor-based safety enhances reliability, ease of use, and makes the system suitable for robotics learning and basic automation applications.

2.3 PROJECT INITIATION

The project was initiated with the aim of designing and developing a Bluetooth Controlled Robotic Car with Obstacle Avoidance capable of performing wireless vehicle control and basic autonomous safety operations. The main goal of the project is to enable real-time control of a robotic car using a smartphone through Bluetooth communication, thereby improving flexibility and ease of operation compared to traditional wired systems. The system is designed to integrate a microcontroller, Bluetooth module, motor driver, and Direct Current (DC) motors to achieve smooth directional movement.

In addition, an ultrasonic sensor is incorporated to continuously detect obstacles and prevent collisions by automatically stopping or redirecting the vehicle when an object is detected within a specific distance. This approach demonstrates the practical application of embedded systems, wireless communication, and sensor-based automation, providing an efficient and low-cost robotic platform for educational learning and intelligent vehicle control experiments.

2.4 SYSTEM SPECIFICATIONS

This section describes the hardware and software requirements necessary for developing the Bluetooth Controlled Robot car with Obstacle Avoidance.

2.4.1 HARDWARE SPECIFICATIONS

The table 2.1 outlines the hardware resources available on the devices used for developing the system. It includes the essential system configuration required for smooth development and execution of the application.

Table 2.1 Hardware Specifications

S.No	Component	Specification / Description
1	Microcontroller	Arduino UNO (ATmega328P, 16 MHz clock speed)
2	Bluetooth Module	HC-05 / HC-06 Bluetooth module (2.4 GHz, Serial communication)
3	Ultrasonic Sensor	HC-SR04 (Range: 2 cm – 400 cm, Accuracy: ± 3 mm)
4	Motor Driver	L298N Dual H-Bridge Motor Driver
5	Motors	DC Geared Motors (6 V / 12 V)
6	Robot Chassis	Two-wheel or four-wheel robot car chassis
7	Power Supply	Battery pack (7.4 V / 9 V / 12 V)
8	Connecting Wires	Jumper wires (Male-to-Male, Male-to-Female)
9	Breadboard / PCB	Breadboard or general purpose PCB
10	User Device	Android smartphone with Bluetooth support

2.4.1 SOFTWARE SPECIFICATIONS

The table 2.2 lists the software specifications used in developing the application, including the operating system, development framework, programming languages, libraries, and Integrated Development Environment (IDE) involved in the project.

Table 2.2 Software Specifications

S.No	Software / Tool	Specification / Description
1	Operating System	Windows / Linux / macOS
2	IDE	Arduino IDE
3	Programming Language	Embedded C / C++
4	Mobile Application	Android Bluetooth control app
5	Communication Protocol	Bluetooth Serial Communication (ASCII commands)
6	Baud Rate	9600 bps
7	Driver Software	USB-to-Serial driver (Arduino USB driver / CH340)
8	Firmware Upload	Arduino Bootloader
9	Diagram Tool	PlantUML / Eraser.io
10	Version Control (Optional)	Git

2.5 TOOLS AND TECHNOLOGIES USED

The tools, libraries, and technologies used in the project are listed below:

2.5.1 Arduino Uno

The Arduino Uno is a microcontroller board based on the ATmega328P microcontroller and serves as the main control unit of the robotic system. It processes input signals from the Bluetooth module and ultrasonic sensor, executes the programmed logic, and sends control signals to the motor driver. The Arduino Uno provides digital and analog input/output pins, making it suitable for interfacing with sensors, communication modules, and actuators. It is programmed using the Arduino IDE with Arduino C/C++.

2.5.2 HC-05 Bluetooth Module

The HC-05 Bluetooth module is used to establish wireless communication between the smartphone and the robotic car. It operates using the Serial Port Protocol (SPP) and allows the Arduino to receive control commands from a mobile device. The module communicates with the Arduino through Universal Asynchronous Receiver Transmitter (UART) serial communication. By pairing the smartphone with the HC-05 module, movement commands such as forward, backward, left, right, and stop can be transmitted to control the robot.

2.5.3 HC-SR04 Ultrasonic Sensor

The HC-SR04 ultrasonic sensor is used for obstacle detection and distance measurement. It works by sending an ultrasonic sound wave through the trigger pin and measuring the time taken for the echo to return after hitting an object. Based on the echo time, the distance between the robot and the obstacle is calculated. This sensor helps the robot identify obstacles in its path and ensures safe operation by preventing collisions.

2.5.4 L298N Motor Driver Module

The L298N motor driver module is a dual H-bridge motor driver used to control the direction and speed of DC motors. It acts as an interface between the Arduino microcontroller and the motors, allowing the Arduino to control motor movement using low-power signals. The module can drive two DC motors simultaneously and supports forward, reverse, left, and right movements of the robotic car.

2.5.5 DC Motor

DC motors are used to provide mechanical movement to the robotic car. These motors convert electrical energy into rotational mechanical motion, which drives the wheels of the robot. By controlling the direction of current through the motors using the motor driver, the robot can move forward, backward, or turn in different directions.

2.5.6 Bluetooth RC Controller

The Bluetooth RC Controller is an Android mobile application used to send control commands to the robotic car via Bluetooth. The application provides a user interface with control buttons such as forward, backward, left, right, and stop. When a user presses a button, the app sends corresponding commands as American Standard Code for Information Interchange (ASCII) characters to the HC-05 Bluetooth module, enabling wireless manual control of the robot.

2.5.7 Arduino C/C++

Arduino C/C++ is the programming language used to develop the control logic for the Arduino microcontroller. It is based on the standard C/C++ programming language and is supported by the Arduino development environment. The program includes functions for initializing hardware components, reading Bluetooth commands, measuring distance using the ultrasonic sensor, and controlling the motor driver to move the robot safely.

CHAPTER 3

SYSTEM DESIGN

System design is the process of crafting the structure and framework of a system to meet specific requirements and objectives. It involves translating user needs and system specifications into a detailed plan that outlines how various components will function together to deliver the desired outcomes. This plan serves as a foundation for development, ensuring that the system is effective, scalable, and user-oriented.

Key aspects of system design include:

- **Logical Design:** Focuses on conceptual frameworks, such as workflows, data relationships, and process diagrams, to define how the system operates at an abstract level.
- **Physical Design:** Deals with tangible elements, including hardware, software, networks, and user interfaces, ensuring the system practical implementation aligns with design goals.
- **Interface Design:** Defines how system components connect and interact, both with each other and with the users, ensuring seamless communication and usability.

System design plays a pivotal role in creating systems that are robust, efficient, and adaptable. It aligns technical capabilities with user expectations and organizational goals, ensuring the system performs reliably under expected condition.

3.1 SYSTEM ARCHITECTURE

System architecture defines the structure and behavior of a system, representing how its components interact and function together to meet specific requirements. System architecture is crucial for ensuring scalability, reliability, and efficiency in system design and implementation.

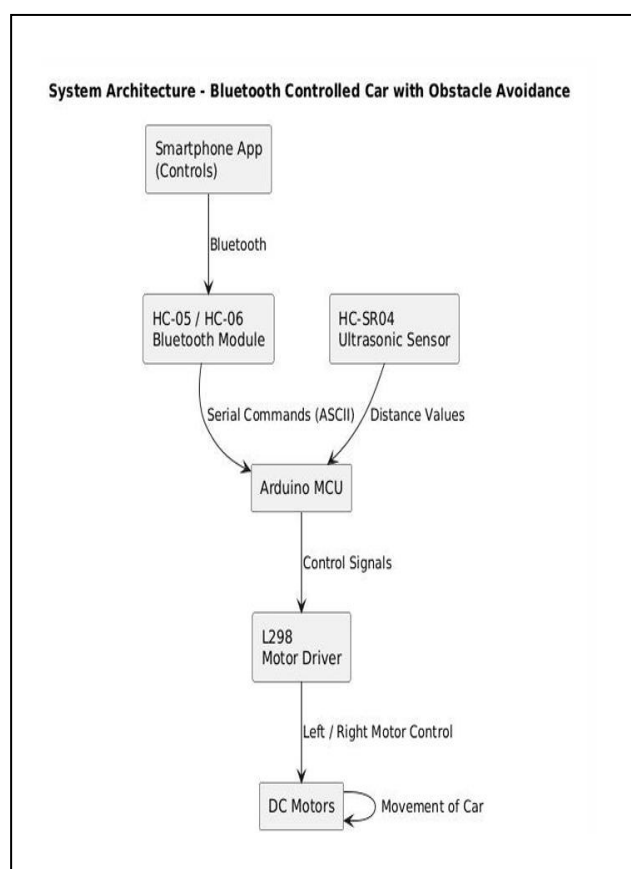


Figure 3.1 System Architecture

Figure 3.2 represents the System Architecture diagram illustrating the interactions between the various components in robot car.

The system architecture of the Bluetooth Controlled Car with Obstacle Avoidance shows the interaction between the user, control modules, sensors, and actuators. The process begins with a smartphone application that sends movement commands via Bluetooth. The Bluetooth module receives these commands and forwards them to the Arduino microcontroller, which acts as the central processing unit.

At the same time, the ultrasonic sensor continuously measures the distance to obstacles and sends data to the Arduino. Based on this data, the Arduino decides whether to execute the user command or stop the robot to avoid collision. It then sends control signals to the motor driver, which drives the Direct Current (DC) motors to perform the required movement. This architecture ensures real-time wireless control along with automatic obstacle detection and safe operation.

3.2 UML DIAGRAMS

Unified Modeling Language (UML) diagrams are a set of standardized visual representations used to model systems, processes, and their components. They help developers and stakeholders understand system requirements and designs. UML diagrams are categorized into structural diagrams and behavioral diagrams (e.g., Use Case, Sequence). This project consists of following UML diagrams for virtual representation of the whole system.

- Use Case Diagram
- Sequence Diagram

3.2.1 Use Case Diagram

A Use Case diagram represents the interactions between users and the system. It highlights system functionalities and the relationships between actors and use cases.

Purpose: To capture system requirements and identify user interactions.

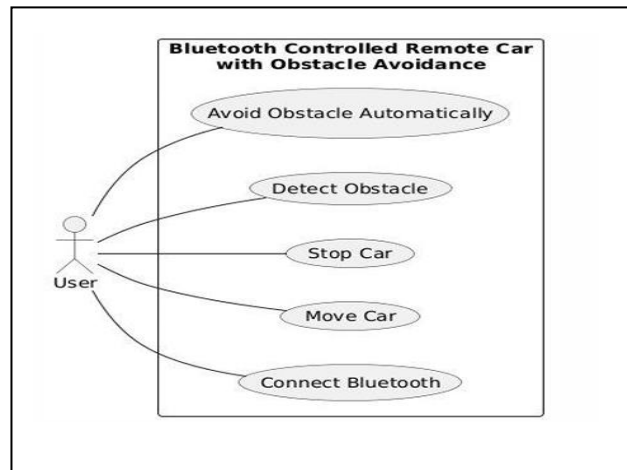


Figure 3.2 Use case Diagram

Figure 3.2 represents the Use Case diagram illustrating the interactions between the user and the robot car.

3.2.2 Sequence Diagram

A sequence diagram illustrates the flow of messages between system components or objects in a specific sequence. It shows how processes interact over time.

Purpose: To model dynamic behaviour and time-based interactions.

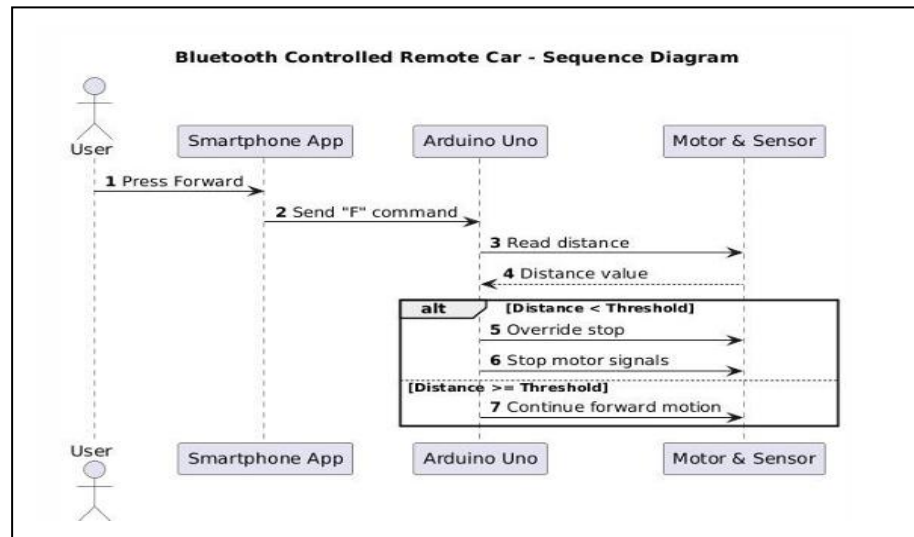


Figure 3.3 Sequence Diagram

Figure 3.3 represents sequence diagram illustrating the flow of data and interactions between the user and robot car.

CHAPTER 4

SYSTEM IMPLEMENTATION AND TESTING

The System implementation refers to the process of integrating a designed system into a functioning environment to ensure it meets predefined requirements and operates effectively. It encompasses installing hardware and software, configuring system settings, conducting testing to verify functionality, training users, and transitioning from the old system to the new one. The goal of system implementation is to deploy the system efficiently, address any issues that arise during the process, and ensure that it delivers the intended outcomes in real-world usage.

4.1 SYSTEM MODULES

A module is a software component or a part of a program that contains one or more routines. One or more independently developed modules make up a program. The project involves the following modules:

- Initialization Module
- Bluetooth Command Module
- Ultrasonic Distance Measurement Module
- Obstacle Detection & Safety Module
- Motor Control Module
- Decision-Making Module (Main Control Logic)
- Continuous Execution Module (loop() Function)
- Android Application Module

4.1.1 Initialization Module

The Initialization Module prepares the Arduino system before the robotic car starts functioning. In this stage, the required pin numbers for the motors, ultrasonic sensor, and Bluetooth module are defined in the program. The Arduino initializes serial communication to establish a connection with the HC-05 Bluetooth module. It also configures the input and output pins required for sensors and motor control. This module ensures that all hardware components are properly configured before the main program begins execution.

4.1.2 Bluetooth Command Module

The Bluetooth Command Module enables wireless communication between the smartphone and the robotic car. It continuously reads incoming data from the HC-05 Bluetooth module using serial communication. The received data is interpreted as movement commands such as forward, backward, left, right, or stop. These commands are processed and passed to the motor control logic for execution. This module allows the user to remotely control the robot using a mobile application.

4.1.3 Ultrasonic Distance Measurement Module

The Ultrasonic Distance Measurement Module is responsible for detecting obstacles in front of the robot. The HC-SR04 ultrasonic sensor sends a trigger pulse and measures the time taken for the echo signal to return after reflecting from an object. The Arduino calculates the distance based on the echo time. This process is repeated continuously to monitor the surroundings. The measured distance helps the system determine whether an obstacle is present in the robot's path.

4.1.4 Obstacle Detection and Safety Module

The Obstacle Detection and Safety Module ensures safe movement of the robotic car by preventing collisions. It compares the distance measured by the ultrasonic sensor with a predefined threshold value. If an obstacle is detected within the threshold distance, the system immediately stops the motors. This safety mechanism overrides any command received from the Bluetooth module. As a result, the robot can avoid obstacles and operate more safely.

4.1.5 Motor Control Module

The Motor Control Module controls the movement of the robotic car using the L298N motor driver. The Arduino sends control signals to the motor driver to regulate the rotation and direction of the DC motors. Depending on the received command, the robot can move forward, backward, turn left, or turn right. The module also handles the stop command when necessary. This module converts electronic control signals into physical motion of the robot.

4.1.6 Decision-Making Module

The Decision-Making Module acts as the central processing logic of the system. It integrates inputs received from the Bluetooth command module and the ultrasonic distance measurement module. The Arduino evaluates whether the robot can safely perform the requested movement. If an obstacle is detected, the safety condition takes priority and the robot stops instead of executing the command. This module ensures proper coordination between user control and automatic safety features.

4.1.7 Continuous Execution Module

The Continuous Execution Module is implemented through the `loop()` function in the Arduino program. This function runs continuously while the system is powered on. It repeatedly checks for new Bluetooth commands and measures the distance using the ultrasonic sensor. Based on these inputs, the system updates the motor actions in real time. This continuous cycle enables responsive and smooth operation of the robotic system.

4.1.8 Android Application Module

The Android Application Module provides the interface for users to control the robotic car through a smartphone. The application establishes a Bluetooth connection with the HC-05 module attached to the robot. It contains control buttons such as forward, backward, left, right, and stop. When a user presses a button, the application sends the corresponding command to the Arduino through Bluetooth communication. This module enables convenient and real-time wireless control of the robot.

4.2 SCREENSHOTS

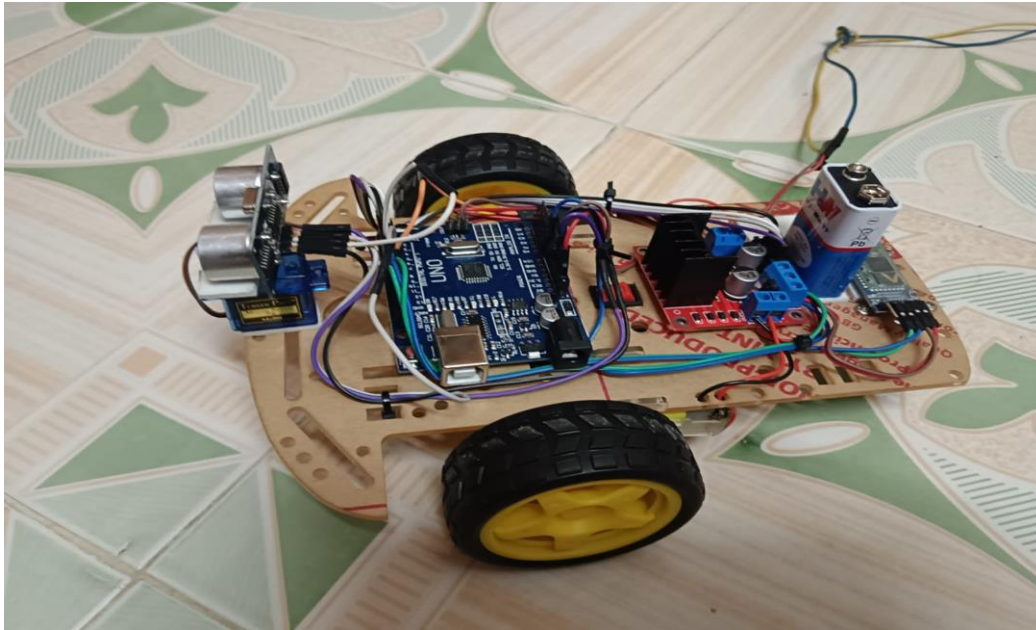


Figure 4.1 Hardware Setup of Bluetooth-Controlled Car

Figure 4.1 shows the physical setup of the robotic car captured using a camera. It includes components like Arduino, motors, ultrasonic sensor, and Bluetooth module integrated together.

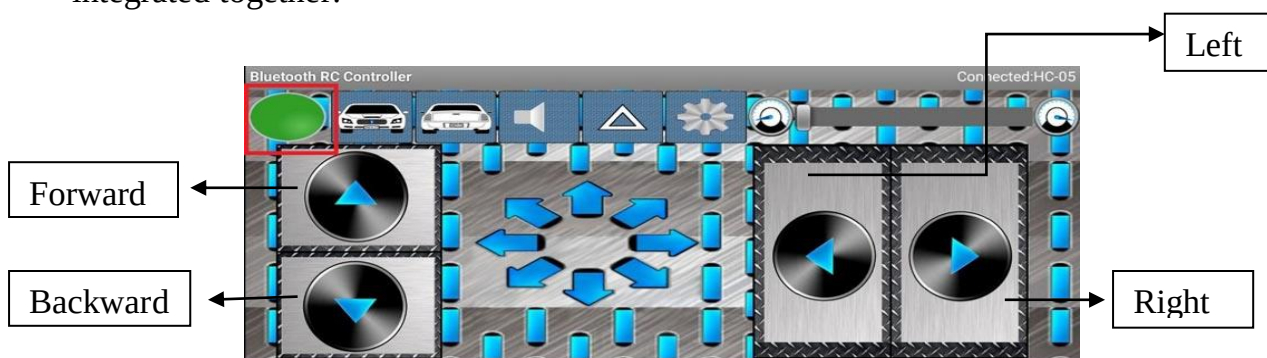


Figure 4.2 Bluetooth Controller Application Interface

Figure 4.2 shows the mobile application used to control the robot via Bluetooth.

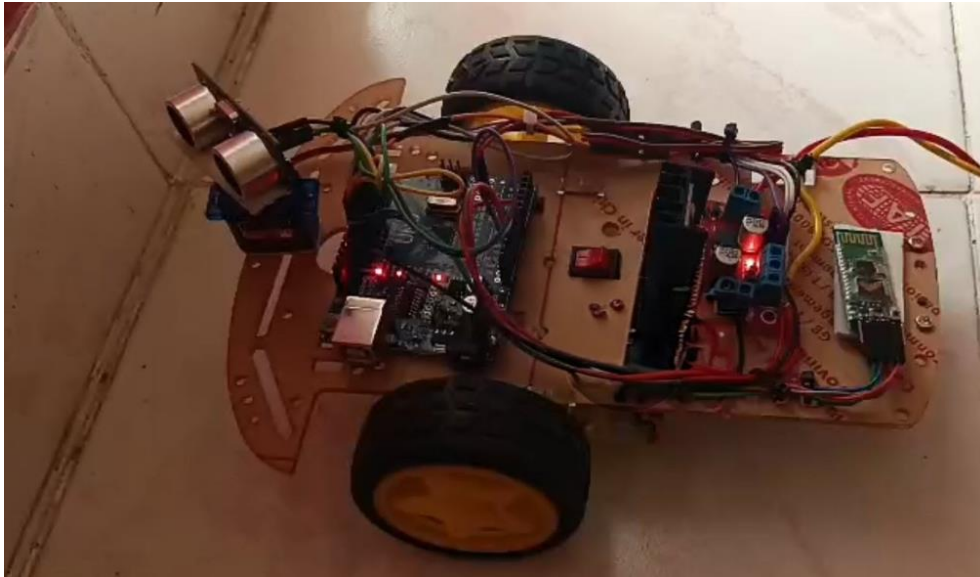


Figure 4.3 Obstacle detected by ultrasonic sensor

Figure 4.3 depicts the movement of ultrasonic sensor searching for a path to operate the car in a collision free environment.

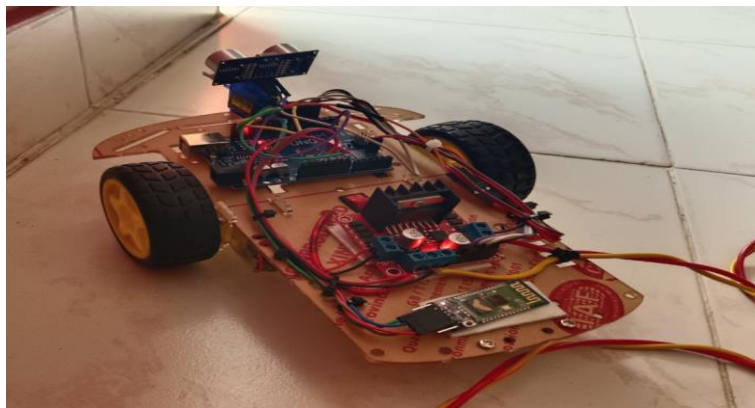


Figure 4.4 Car avoiding the obstacle

Figure 4.4 depicts the movement of car to the another direction to avoid the obstacle

4.3 TESTING

Software and hardware testing is the process of evaluating a system to ensure that it functions according to the specified requirements and performs reliably under real-world conditions. The primary goal of testing is to identify defects, verify system behavior, and ensure the stability of the integrated components before final deployment. For validating the functionality, reliability, and performance of the Bluetooth Controlled Robot Car with Obstacle Avoidance system, several testing strategies were applied including unit testing, system testing, integration testing, performance testing, and user acceptance testing.

4.3.1 Unit Testing

Unit Testing focuses on testing individual components of the system independently to ensure that each functional unit operates correctly. This approach helps detect errors in the early stages of development and improves overall system reliability.

- **Bluetooth Communication Module** - Unit testing of the Bluetooth module verifies the ability of the HC-05 module to establish a connection with the smartphone application. The test ensures correct transmission and reception of command characters such as forward, backward, left, right, and stop. It also checks for communication delays, signal stability, and proper serial data handling by the Arduino controller.

- **Ultrasonic Sensor Module** - Unit testing for the ultrasonic sensor verifies accurate distance measurement using the HC-SR04 sensor. The tests ensure correct trigger pulse generation, echo signal detection, and distance calculation based on echo time. The module is also tested for proper response when objects are placed at different distances from the robot.

- **Motor Control Module** - Unit testing of the motor control module ensures that the L298N motor driver correctly responds to control signals from the Arduino. Each movement command is tested individually to verify forward movement, backward movement, left turn, right turn, and stop operations. The tests confirm proper direction control and stable motor operation.
- **Obstacle Detection and Safety Module** - Unit testing validates the safety logic that prevents collisions. The module checks whether the system correctly compares measured distance with the threshold value. If an obstacle is detected within the defined range, the robot must stop immediately regardless of user command.
- **Android Application Module** - Unit testing verifies that the mobile application correctly sends control commands through Bluetooth. Each button on the interface is tested to ensure that the correct command signal is transmitted and received by the Arduino controller.

4.3.2 System Testing

System Testing evaluates the complete robotic system as a whole to ensure that all components function together correctly under real-time conditions.

- **Bluetooth Communication and Robot Control** - System testing verifies that the smartphone application successfully connects to the HC-05 module and sends commands to the robot. The robot should respond immediately to each command without delay or communication failure.
- **Ultrasonic Obstacle Detection System** - System tests confirm that the ultrasonic sensor accurately detects obstacles in real-time while the robot is moving. The robot must automatically stop when an obstacle appears within the safety distance.

- **Motor Movement and Direction Control** - Testing verifies that the robot performs correct movement operations such as forward, backward, left, and right turns under different operating conditions.
- **Safety Override Mechanism** - The system testing ensures that obstacle detection overrides Bluetooth commands when a collision risk is detected. This guarantees safe operation even when the user continues to send movement commands.
- **Continuous Operation Stability** - Tests are conducted to ensure that the robot operates continuously without system crashes, power instability, or communication interruptions.

4.3.3 Integration Testing

Integration Testing verifies the correct interaction and data flow between different modules once individual components have been tested.

- **Bluetooth Module and Motor Control Module** - Integration testing ensures that commands received from the Bluetooth module are correctly interpreted by the Arduino and executed by the motor control system.
- **Ultrasonic Sensor and Safety Control Module** - Tests verify that distance measurements from the ultrasonic sensor are correctly passed to the safety logic for obstacle detection and motor control decisions.
- **Sensor System and Motor Control and Decision Logic** - Integration testing confirms that the robot simultaneously processes sensor data and user commands while maintaining safe navigation.

4.3.4 Performance Testing

Performance Testing evaluates the efficiency and responsiveness of the robotic system under continuous operation. Metrics such as command response time, obstacle detection accuracy, and motor performance are analyzed. The robot is tested under repeated command inputs and varying obstacle conditions to ensure reliable performance during extended operation.

4.4 TEST CASES

A test case is a structured set of instructions used to verify whether a particular feature or functionality of a system operates as intended. It specifies the input conditions, execution steps, expected results, and actual results observed during testing. Test cases help ensure that every component of the system behaves correctly and that the overall system meets its design requirements.

In the **Bluetooth Controlled Robot Car with Obstacle Avoidance system**, test cases are designed to validate the correct operation of hardware components such as the Bluetooth module, ultrasonic sensor, motor driver, and the Arduino controller, as well as the interaction between the mobile application and the robotic system.

Below are some major reasons for developing test cases for this project:

- **Verify correct functioning of each module** – Test cases ensure that individual components such as the Bluetooth communication module, ultrasonic sensor, and motor control system work properly under different operating conditions.
- **Detect faults during development** – Writing and executing test cases helps identify errors in hardware connections, sensor readings, and software logic early in the development stage, reducing the complexity of fixing issues later.

- **Provide clear system documentation** – Test cases serve as a reference that explains how different parts of the robotic system operate and how they should behave when specific commands or conditions occur.
- **Promote modular and maintainable design** – By testing each subsystem independently, developers are encouraged to design the system in smaller, manageable modules that are easier to maintain and upgrade.
- **Support regression testing** – When modifications are made to the system, existing test cases can be reused to confirm that new changes do not negatively affect previously working functionalities.
- **Ensure reliable real-world operation** – Through proper testing, the robot can be verified to respond accurately to Bluetooth commands and safely stop when obstacles are detected, ensuring safe and reliable operation for users.

Table 4.1 Test Cases

Module	Test Case Summary	Expected Result	Actual Result	Status (Pass/Fail)
Bluetooth Communication Module	Verify Bluetooth pairing between smartphone and HC-05 module	Successful pairing and stable Bluetooth connection established	Connection established successfully and remained stable	Pass
Motor Control Module	Test forward movement command from mobile application	Robot moves forward when no obstacle is detected	Robot moved forward correctly without interruption	Pass
Direction Control Module	Test left and right movement commands	Robot turns correctly in the selected direction	Robot changed direction accurately as commanded	Pass

Table 4.1 Test Cases (Continued)

Motor Control Module	Verify stop command functionality	Robot stops immediately when stop command is issued	Robot stopped instantly upon receiving command	Pass
Obstacle Detection Module	Detect obstacle using ultrasonic sensor within threshold distance	Robot automatically stops when obstacle is detected	Robot stopped automatically when object was placed in front	Pass
Safety Control Module	Verify obstacle safety override while forward command is active	Robot should not move forward when obstacle is present	System prevented forward movement as expected	Pass
System Integration Module	Test continuous command execution during operation	Robot should respond to multiple commands without delay or malfunction	Robot responded to all commands smoothly without lag	Pass

CHAPTER 5

CONCLUSION

This project successfully implements a Bluetooth Controlled Robot Car with Obstacle Avoidance, enabling remote navigation with built-in safety features. By integrating the Arduino microcontroller, HC-05 Bluetooth module, ultrasonic sensor, motor driver, and Direct Current (DC) motors, the system allows users to control the robot in real time using a smartphone. The ultrasonic sensor detects obstacles and automatically stops the robot to prevent collisions, ensuring safe operation.

The project demonstrates effective integration of hardware and software components, providing reliable communication, accurate sensing, and stable movement control. Its modular design allows easy future enhancements such as improved navigation, additional sensors, and autonomous capabilities. Overall, the system presents a practical and scalable solution for wireless robotic control and highlights the potential of embedded systems in modern robotics applications.

5.1 LIMITATIONS

- **Limited Operational Range and Control Dependency** - The system depends on Bluetooth communication for control, which restricts the operational range of the robot. The robot cannot be controlled beyond a short distance and requires continuous user input for movement, limiting its usability in larger environments.
- **Basic Obstacle Avoidance Mechanism** – The project implements only simple front-side obstacle detection and stopping functionality. It does not support advanced navigation features such as path planning, obstacle avoidance from multiple directions, or intelligent rerouting.

- **Lack of Real-Time Monitoring and Feedback** – The robot does not provide real-time feedback such as video streaming or detailed environment data to the user. This limits the user's ability to monitor surroundings and control the robot effectively in complex or remote scenarios.

5.2 FUTURE ENHANCEMENTS

- **Wi-Fi Based Control** – The robot can be upgraded to use Wi-Fi or IoT technology, allowing it to be controlled from anywhere through the internet instead of limited Bluetooth range.
- **Camera Integration** – A camera module can be added to provide live video streaming for surveillance and remote monitoring purposes.
- **Advanced Autonomous Navigation** – Intelligent algorithms and additional sensors can be implemented to enable the robot to navigate automatically without manual control.

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